



Problem Statement 02: Predicting Hotel Booking Cancellations & Revenue Risk

Business Scenario

A hotel group manages properties across different locations and receives thousands of bookings every year.

While the hotel has access to detailed booking information, management currently has limited ability to **predict which bookings are likely to be cancelled** and understand the potential revenue impact of those cancellations.

Last-minute cancellations can result in:

- Empty rooms
- Lost revenue
- Inefficient resource planning
- Difficulty in managing occupancy
- Increased dependency on last-minute bookings

Management wants to move from simply analyzing historical booking data to **predicting future booking outcomes**.

Challenge

Using the provided hotel booking dataset, **build a machine learning solution that predicts whether a hotel booking is likely to be cancelled and provides insights into the factors driving cancellation risk.**

The solution should help hotel management identify potentially risky bookings and make better decisions around:

- Occupancy planning
- Customer retention
- Revenue management
- Booking strategies

Teams should focus on building one strong predictive solution rather than attempting multiple unrelated models.

Key Data Science Questions

Participants should investigate questions such as:

1. Which booking characteristics are strong predictors of cancellation?
2. Can we predict whether a new booking will be cancelled?
3. Which customer segments have the highest cancellation risk?
4. How do lead time, deposit type, market segment, previous cancellations, and other booking characteristics affect cancellation probability?
5. Which features contribute most strongly to the model's predictions?
6. How could the hotel use cancellation predictions to reduce revenue loss?

Dataset

Hotel Booking Demand Dataset

The dataset contains approximately **119,390 hotel bookings** and includes information about booking behavior, customers, stay duration, market segments, and reservation outcomes.

Important Features Include:

- Hotel type
- Lead time
- Arrival date
- Length of stay
- Number of guests
- Meal type
- Country
- Market segment
- Distribution channel
- Repeated guest status
- Previous cancellations
- Previous bookings
- Reserved room type
- Assigned room type
- Booking changes
- Deposit type
- Customer type
- Average Daily Rate (ADR)
- Special requests
- Reservation status

Target Variable

`is_cancelled`

0 → Booking was not cancelled
1 → Booking was cancelled



Expected Data Science Workflow

1. Exploratory Data Analysis

Explore the dataset to understand:

- Booking patterns
- Cancellation patterns
- Customer behavior
- Seasonal trends
- Market segments
- Stay duration
- Pricing patterns

Identify relationships that could be useful for predicting cancellations.



2. Data Preparation

Participants should:

- Handle missing values
 - Identify inconsistencies
 - Handle categorical variables
 - Remove irrelevant features where appropriate
 - Check for potential data leakage
 - Prepare the dataset for machine learning
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3. Feature Engineering

Participants are encouraged to create meaningful features from the existing data.

Examples may include:

- Total stay duration
- Total number of guests
- Booking-to-arrival intervals
- Previous cancellation behavior
- Customer history
- Seasonal features
- Price-related features

Teams should explain why the engineered features may help the model.



4. Model Development

Build a **classification model** capable of predicting whether a booking will be cancelled.

Teams may experiment with appropriate algorithms such as:

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting
- XGBoost / LightGBM
- Other suitable classification approaches

No specific machine learning algorithm is mandatory.

Teams should justify their model choice.



5. Model Evaluation

Evaluate the model using appropriate classification metrics.

Possible metrics include:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC
- Confusion Matrix

Since identifying potential cancellations is important to the business, teams should explain the **trade-off between false positives and false negatives**.

6. Model Interpretability

Management needs to understand **why the model considers a booking risky**.

Participants should identify:

- Important features
 - Factors associated with cancellation
 - High-risk booking characteristics
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Business Recommendations

Based on the analysis and model results, provide at least **3 actionable recommendations**.

For example:

"Bookings with long lead times and certain deposit types have a significantly higher cancellation probability."

Possible business actions could include:

- Targeted confirmation reminders
- Flexible cancellation policies
- Deposit strategies

- Customer retention offers
- Dynamic overbooking strategies
- Targeted marketing

Recommendations should be supported by **data and model findings**.

Expected Outcome

The final solution should help the hotel answer:

| **"Which bookings are likely to be cancelled?"**

and:

| **"Why are they likely to be cancelled?"**

and ultimately:

| **"What should the hotel do about it?"**

The goal is to transform historical booking data into a **predictive decision-support solution**.

Required Deliverables

1. Data Science Notebook

The notebook should include:

- Data Understanding
- Data Cleaning
- Exploratory Data Analysis
- Feature Engineering
- Model Development
- Model Evaluation

- Model Interpretation
 - Business Recommendations
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2. Trained Machine Learning Model

Submit:

- Trained model, or
- Reproducible model-training code

The solution should be capable of making predictions on unseen booking records.



3. Prediction & Insights

Provide visualizations showing:

- Cancellation patterns
 - Model performance
 - Confusion matrix
 - Feature importance
 - Key factors affecting cancellation risk
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PPT Presentation

Maximum 5 Slides

1. Business Understanding

- Business problem
- Why cancellation prediction matters
- Selected approach

2. Data & Key Insights

- Dataset overview
- EDA findings
- Important features

3. Modeling Approach

- Feature engineering
- Models tested
- Selected model

4. Results & Interpretation

- Model performance
- Confusion matrix
- Important features
- Example predictions

5. Recommendations & Business Impact

- Actionable recommendations
- Expected business impact
- Future improvements



Time Limit

4 Hours

Teams should focus on building a **working end-to-end predictive solution**.

A well-evaluated model solving one meaningful business problem is better than multiple incomplete models.



AI Usage

AI IS ALLOWED

Participants may use AI tools such as:

- ChatGPT
- Claude
- Gemini
- GitHub Copilot
- Other AI-assisted development tools

AI may be used for:

- Coding
- Debugging
- EDA assistance
- Feature engineering ideas
- Model selection
- Visualization
- Documentation
- Presentation preparation

However:

Teams are responsible for understanding, validating, and explaining their solution.

Judges may ask:

- Why did you choose this target?
- Why did you choose this model?
- Which evaluation metric matters most?
- What causes the model to predict cancellation?

- How would the hotel use your predictions?
 - How did you avoid data leakage?
 - What are the limitations of your model?
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Evaluation Criteria

Criteria	Weight
 Problem Understanding	15%
 EDA & Feature Engineering	20%
 Model Development	25%
 Evaluation & Interpretability	15%
 Business Recommendations	15%
 Presentation & Demo	10%
Total	100%
